HyperFitPro Studio
Hyperelastic Material Modeling and Calibration Theory Manual

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Mechanical modeling, stress engineering, and material characterization

Preparation of hyperelastic material model inputs for finite element simulations

Document purpose

This document explains the mechanical basis of hyperelastic material models, how stress response is obtained from test data, how parameters should be interpreted, and which checks are required during calibration. The goal is not only to list equations, but to make model selection, parameter influence, validity limits and engineering reporting checks clear.

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1 General Approach

1.1 Meaning of calibration

Hyperelastic calibration is broader than fitting a mathematical curve with the lowest residual error. The physical meaning of the model, adequacy of test modes, identifiability of parameters and limiting behavior should be evaluated together.

Basic decision rule

The model with the lowest error is not always the most reliable model. A reliable model captures the initial slope, remains physical at high deformation, behaves consistently across deformation modes and does not move its parameters into invalid regions.

1.2 Engineering check levels

Check level	Topic	Interpretation
Mechanical level	Energy function and model family	Checks whether the model is consistent with expected material behavior.
Data level	Test-mode coverage	Uniaxial data alone is not sufficient for many models; biaxial, planar and shear data separate parameters.
Numerical level	Optimization stability	Checks whether parameters are stuck on bounds and whether the curve oscillates.
Reporting level	Validity range	Defines the stretch and stress range in which the model may be used.

Table 1.1: Core check levels for hyperelastic model calibration.

2 Kinematic Basis

2.1 Deformation measures

$$\boldsymbol{F} = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}}, \qquad J = \det \boldsymbol{F}, \qquad \boldsymbol{C} = \boldsymbol{F}^T \boldsymbol{F}$$

For incompressible response,

$$\lambda_1 \lambda_2 \lambda_3 = 1.$$

The first and second invariants are

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2, \qquad I_2 = \lambda_1^2\lambda_2^2 + \lambda_2^2\lambda_3^2 + \lambda_3^2\lambda_1^2.$$

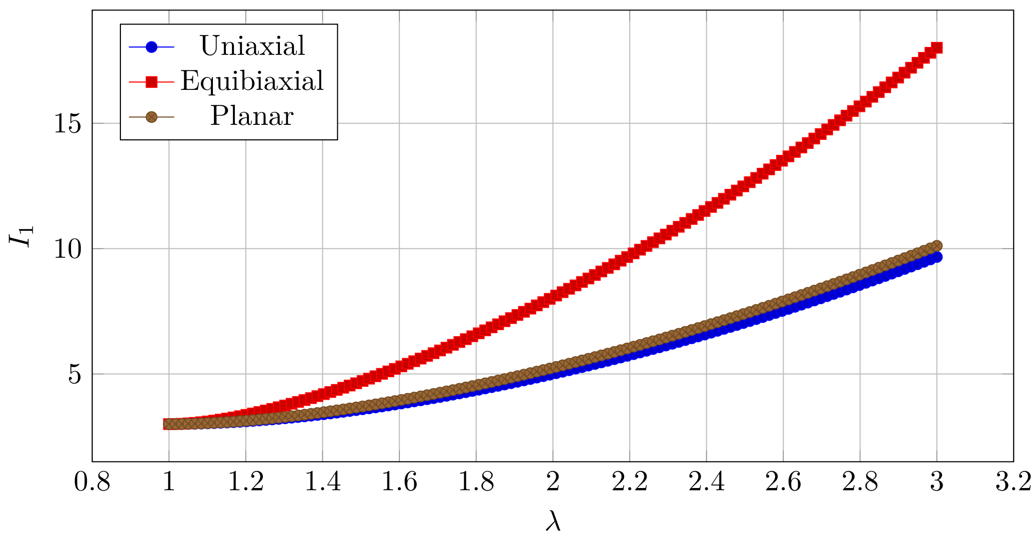

Figure 2.1: Variation of I_1 for different test modes at the same stretch.

3 Stress Response by Test Mode

3.1 Core equations

For incompressible invariant-based models,

$$W_1 = \frac{\partial W}{\partial I_1}, \qquad W_2 = \frac{\partial W}{\partial I_2}.$$

Test mode	Principal stretches	Nominal or shear stress expression
Uniaxial	$\lambda_1 = \lambda,\ \lambda_2 = \lambda_3 = \lambda^{-1/2}$	$P = 2(\lambda - \lambda^{-2})(W_1 + \lambda^{-1} W_2)$
Equibiaxial	$\lambda_1 = \lambda_2 = \lambda,\ \lambda_3 = \lambda^{-2}$	$P = 2(\lambda - \lambda^{-5})(W_1 + \lambda^{2} W_2)$
Planar	$\lambda_1 = \lambda,\ \lambda_2 = 1,\ \lambda_3 = \lambda^{-1}$	$P = 2(\lambda - \lambda^{-3})(W_1 + W_2)$
Simple shear	$I_1 = I_2 = 3 + \gamma^2$	$\tau_{12} = 2\gamma(W_1 + W_2)$

Table 3.1: Test-mode equations for incompressible invariant-based models.

4 Model Family Comparison

Model family	Data need	Strength	Main caution
Neo-Hookean	Initial uniaxial check	Simple, stable and fast	Limited high-strain stiffening
Mooney-Rivlin	Uniaxial plus biaxial or planar data	Uses I_1 and I_2 contributions	Parameters may not separate from one mode
Yeoh / reduced polynomial	Fast with uniaxial data	Practical for filled rubber candidates	Does not directly represent I_2 effects
Ogden	Multiple test modes	Flexible in tension and compression	Parameters may not be unique
Gent / Arruda-Boyce	High-stretch data	Represents finite chain extensibility	Limit parameter must not approach the test range
Exponential models	High-deformation data	Captures strong stiffening	Wide bounds can create artificial steepening

Table 4.1: Engineering comparison of hyperelastic model families.

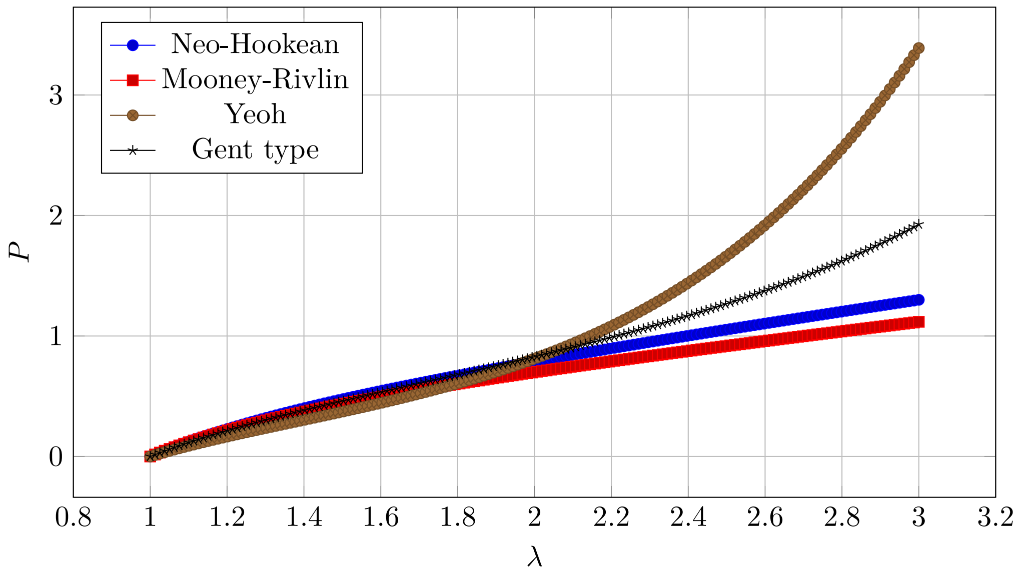

Figure 4.1: Example uniaxial nominal stress curves for different model families.

4.1 How to read parameter influence degree

The parameter influence chart in each model card shows how strongly a parameter changes the model response. It is not a material constant; it is a relative indicator used to communicate calibration sensitivity.

Influence degree	Meaning
1/5	Weak or auxiliary effect on model response
2/5	Secondary shape correction
3/5	Noticeable change in curve shape or secondary mode contribution
4/5	Strong effect on high-stretch, coupling or exponent behavior
5/5	Critical effect on initial stiffness, limit safety or exponential stiffening

Table 4.2: Interpretation of parameter influence degree.

5 Limit Parameters and Invalid-Zone Control

Gent, Arruda-Boyce and similar models contain limit parameters such as I_L, J_L or λ_L. These parameters represent that the material cannot stretch indefinitely. If the limit is too close to the last data point, the fitted curve may look accurate but can become nonphysical immediately outside the calibration range.

Readable technical check

For a Gent-type model,

$$q = \frac{I_1 - 3}{I_L - 3}$$

can be calculated. As q approaches 1, the model enters its limiting zone. The report should therefore show not only the error value, but also the largest sampled I_1, the fitted I_L and the safety distance between them.

Limit ratio	Engineering interpretation
$q < 0.70$	Limit is sufficiently away from the test range.
$0.70 \leq q < 0.85$	Usable, but high-stretch slope must be reviewed.
$0.85 \leq q < 0.95$	Limit is close to the test range; use only inside the calibration range.
$q \geq 0.95$	Model is attached to the limiting zone and is not reliable for physical interpretation.

Table 5.1: Validity check table for limiting parameters.

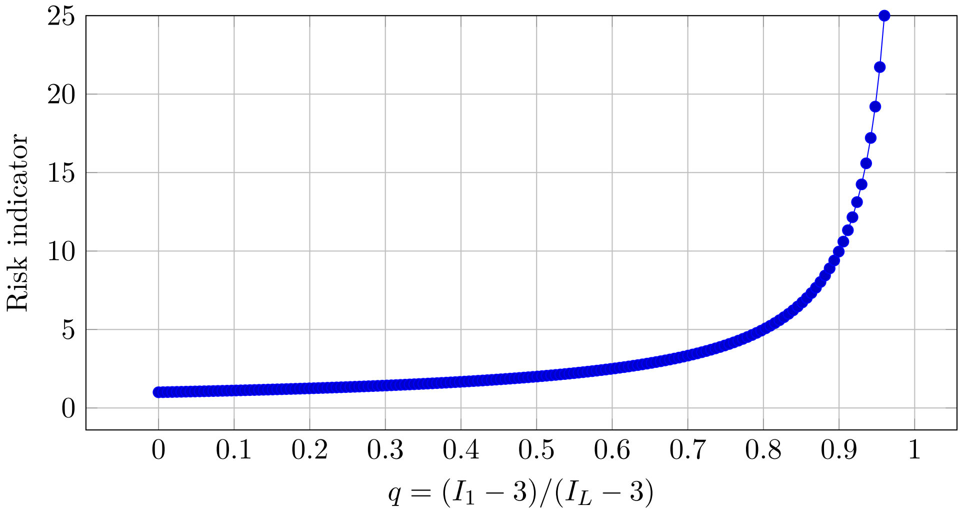

Figure 5.1: Invalid-zone risk increasing as the limit ratio approaches 1.

6 Model Library and Parameter Cards

The bounds in this chapter are default search ranges, not universal material limits. Users should adjust them according to stress unit, stress level, maximum stretch and material class.

6.1 Model 1: 2 term Mooney-Rivlin

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar

6.1.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

6.1.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	-10	10	0.05	stress	-
C_{01}	-10	10	0.05	stress	-

Table 6.1: Model parameter search ranges.

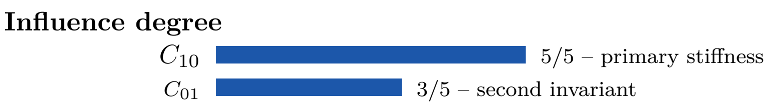

Figure 6.1: Relative influence degree of model parameters on the material response.

6.1.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.2 Model 2: 3 term Mooney-Rivlin

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar

6.2.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) + C_{11}(I_1 - 3)(I_2 - 3)$$

6.2.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	-10	10	0.05	stress	-
C_{01}	-10	10	0.05	stress	-
C_{11}	-10	10	0	stress	-

Table 6.2: Model parameter search ranges.

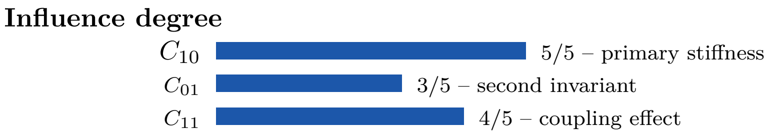

Figure 6.2: Relative influence degree of model parameters on the material response.

6.2.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.3 Model 3: 5 term Mooney-Rivlin

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar, simple shear

6.3.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) + C_{20}(I_1 - 3)^2 + C_{11}(I_1 - 3)(I_2 - 3)$$

6.3.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	-10	10	0.05	stress	-
C_{01}	-10	10	0.05	stress	-
C_{20}	-10	10	0	stress	-
C_{11}	-10	10	0	stress	-

Table 6.3: Model parameter search ranges.

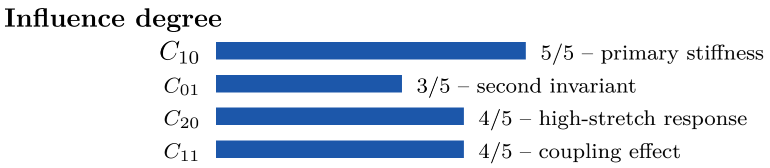

Figure 6.3: Relative influence degree of model parameters on the material response.

6.3.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.4 Model 4: 9 term Mooney-Rivlin

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar, simple shear

6.4.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3 + C_{11}(I_1 - 3)(I_2 - 3)$$

6.4.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	-10	10	0.05	stress	-
C_{01}	-10	10	0.05	stress	-
C_{20}	-10	10	0	stress	-
C_{30}	-10	10	0	stress	-
C_{11}	-10	10	0	stress	-

Table 6.4: Model parameter search ranges.

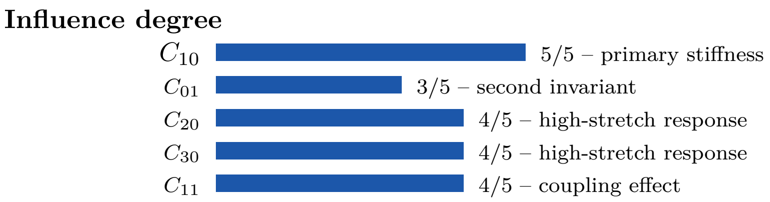

Figure 6.4: Relative influence degree of model parameters on the material response.

6.4.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.5 Model 5: Third degree polynomial

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar, simple shear

6.5.1 Strain-energy function

$$W = C_{10}(I_1-3)+C_{01}(I_2-3)+C_{11}(I_1-3)(I_2-3)+C_{20}(I_1-3)^2+C_{02}(I_2-3)^2+C_{21}(I_1-3)^2(I_2-3)+C_{12}(I_1-3)(I_2-3)^2$$

6.5.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	-10	10	0.05	stress	-
C_{01}	-10	10	0.05	stress	-
C_{11}	-10	10	0	stress	-
C_{20}	-10	10	0	stress	-
C_{02}	-10	10	0	stress	-
C_{21}	-10	10	0	stress	-
C_{12}	-10	10	0	stress	-

Table 6.5: Model parameter search ranges.

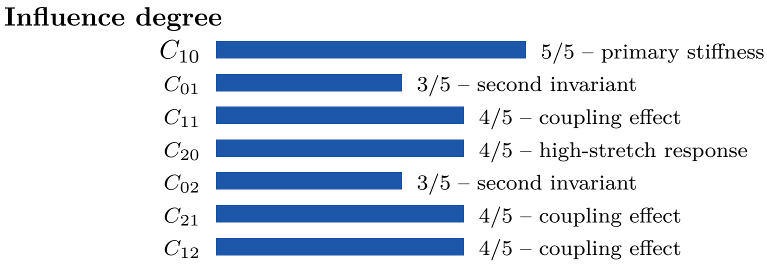

Figure 6.5: Relative influence degree of model parameters on the material response.

6.5.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.6 Model 6: Neo-Hookean

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar

6.6.1 Strain-energy function

$$W = \frac{\mu}{2}(I_1 - 3)$$

6.6.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-

Table 6.6: Model parameter search ranges.

Figure 6.6: Relative influence degree of model parameters on the material response.

6.6.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.7 Model 7: 2 term Yeoh

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar

6.7.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2$$

6.7.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0	20	0.05	positive stress	-
C_{20}	-10	10	0	stress	-

Table 6.7: Model parameter search ranges.

Figure 6.7: Relative influence degree of model parameters on the material response.

6.7.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.8 Model 8: 3 term Yeoh

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar, simple shear

6.8.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3$$

6.8.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0	20	0.05	positive stress	-
C_{20}	-10	10	0	stress	-
C_{30}	-10	10	0	stress	-

Table 6.8: Model parameter search ranges.

Figure 6.8: Relative influence degree of model parameters on the material response.

6.8.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.9 Model 9: 5 term Yeoh

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar, simple shear

6.9.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3 + C_{40}(I_1 - 3)^4 + C_{50}(I_1 - 3)^5$$

6.9.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0	20	0.05	positive stress	-
C_{20}	-10	10	0	stress	-
C_{30}	-10	10	0	stress	-
C_{40}	-10	10	0	stress	-
C_{50}	-10	10	0	stress	-

Table 6.9: Model parameter search ranges.

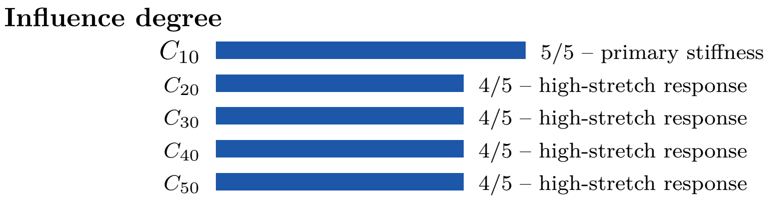

Figure 6.9: Relative influence degree of model parameters on the material response.

6.9.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.10 Model 10: 5 term Arruda-Boyce

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar

6.10.1 Strain-energy function

$$W = \mu \sum_{i=1}^{5} \frac{c_i}{\lambda_L^{2i-2}} (I_1^i - 3^i)$$

6.10.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
λ_L	1.01	20	5	uzama limiti	-

Table 6.10: Model parameter search ranges.

Figure 6.10: Relative influence degree of model parameters on the material response.

6.10.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.11 Model 11: Gent

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar

6.11.1 Strain-energy function

$$W = -\frac{\mu}{2}(I_L - 3)\ln\left(1 - \frac{I_1 - 3}{I_L - 3}\right)$$

6.11.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
I_L	3.01	1000	50	değişmez limiti	-

Table 6.11: Model parameter search ranges.

Figure 6.11: Relative influence degree of model parameters on the material response.

6.11.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.12 Model 12: 2 term Ogden

Model summary

Model family: Principal-stretch / spectral family
Recommended data: uniaxial, biaxial, planar, simple shear

6.12.1 Strain-energy function

$$W = \sum_{i=1}^{2} \frac{\mu_i}{\alpha_i}(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3)$$

6.12.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ_1	-20	20	0.1	stress	-
α_1	-20	20	2	dimensionless	-
μ_2	-20	20	0	stress	-
α_2	-20	20	-2	dimensionless	-

Table 6.12: Model parameter search ranges.

Figure 6.12: Relative influence degree of model parameters on the material response.

6.12.3 Engineering interpretation

This model is directly driven by principal stretches. It can represent tension and compression flexibly, but stress-scale parameters and exponent parameters may compensate each other. Parameters obtained from uniaxial data only should not be treated as unique.

6.13 Model 13: 3 term Ogden

Model summary

Model family: Principal-stretch / spectral family
Recommended data: uniaxial, biaxial, planar, simple shear

6.13.1 Strain-energy function

$$W = \sum_{i=1}^{3} \frac{\mu_i}{\alpha_i}(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3)$$

6.13.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ_1	-20	20	0.1	stress	-
α_1	-20	20	2	dimensionless	-
μ_2	-20	20	0	stress	-
α_2	-20	20	-2	dimensionless	-
μ_3	-20	20	0	stress	-
α_3	-20	20	4	dimensionless	-

Table 6.13: Model parameter search ranges.

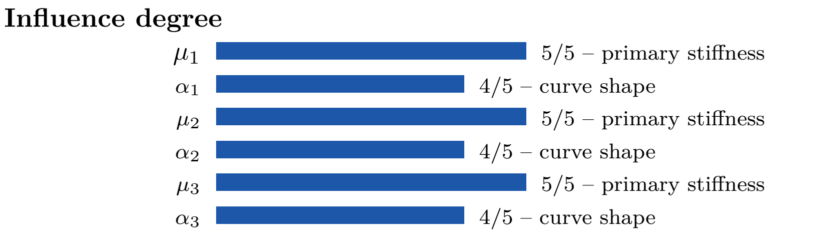

Figure 6.13: Relative influence degree of model parameters on the material response.

6.13.3 Engineering interpretation

This model is directly driven by principal stretches. It can represent tension and compression flexibly, but stress-scale parameters and exponent parameters may compensate each other. Parameters obtained from uniaxial data only should not be treated as unique.

6.14 Model 14: Veronda-Westmann

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar

6.14.1 Strain-energy function

$$W = C_1(e^{\alpha(I_1-3)} - 1) - C_2(I_2 - 3)$$

6.14.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_1	0	20	0.05	positive stress	-
α	1e-08	100	1	dimensionless	-
C_2	-20	20	0	stress	-

Table 6.14: Model parameter search ranges.

Figure 6.14: Relative influence degree of model parameters on the material response.

6.14.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.15 Model 15: Humphrey-Yin

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar

6.15.1 Strain-energy function

$$W = C_1(e^{C_2(I_1-3)} - 1)$$

6.15.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_1	0	20	0.05	positive stress	-
C_2	1e-08	100	1	dimensionless	-

Table 6.15: Model parameter search ranges.

Figure 6.15: Relative influence degree of model parameters on the material response.

6.15.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.16 Model 16: Hart-Smith

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar

6.16.1 Strain-energy function

$$W = \frac{C_1 e^{C_3(I_1-3)^2}}{2} + 3C_2 \ln I_2$$

6.16.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_1	0	20	0.05	positive stress	-
C_2	-20	20	0.05	stress	-
C_3	1e-08	100	1	dimensionless	-

Table 6.16: Model parameter search ranges.

Figure 6.16: Relative influence degree of model parameters on the material response.

6.16.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.17 Model 17: Peng-Landel

Model summary

Model family: Principal-stretch / spectral family
Recommended data: uniaxial, biaxial, planar, simple shear

6.17.1 Strain-energy function

$$W = C_1 \sum_{i=1}^{3} [\lambda_i^2 - 1 - \ln \lambda_i^2 - \frac{1}{6}(\ln \lambda_i^2)^2 + \frac{1}{18}(\ln \lambda_i^2)^3 - \frac{1}{216}(\ln \lambda_i^2)^4]$$

6.17.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_1	0	20	0.05	positive stress	-

Table 6.17: Model parameter search ranges.

Figure 6.17: Relative influence degree of model parameters on the material response.

6.17.3 Engineering interpretation

This model is directly driven by principal stretches. It can represent tension and compression flexibly, but stress-scale parameters and exponent parameters may compensate each other. Parameters obtained from uniaxial data only should not be treated as unique.

6.18 Model 18: Knowles

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar

6.18.1 Strain-energy function

$$W = \frac{\mu}{2b}[(1 + \frac{b(I_1 - 3)}{n})^n - 1]$$

6.18.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
b	1e-08	100	1	dimensionless	-
n	0.05	20	2	dimensionless	-

Table 6.18: Model parameter search ranges.

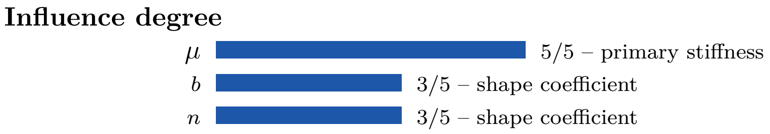

Figure 6.18: Relative influence degree of model parameters on the material response.

6.18.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.19 Model 19: Martins

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar

6.19.1 Strain-energy function

$$W = C_1(e^{C_2(I_1-3)} - 1) + C_3(e^{C_4(\lambda_f-1)^2} - 1)$$

6.19.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_1	0	20	0.05	positive stress	-
C_2	1e-08	100	1	dimensionless	-
C_3	0	20	0.05	positive stress	-
C_4	1e-08	100	1	dimensionless	-

Table 6.19: Model parameter search ranges.

Figure 6.19: Relative influence degree of model parameters on the material response.

6.19.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.20 Model 20: Pucci-Saccomandi

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar, simple shear

6.20.1 Strain-energy function

$$W = -\frac{1}{2}\mu J_L \ln(1 - \frac{I_1 - 3}{J_L}) + C_2 \ln(\frac{1}{3} I_2)$$

6.20.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
J_L	3.01	1000	50	değişmez limiti	-
C_2	-20	20	0	stress	-

Table 6.20: Model parameter search ranges.

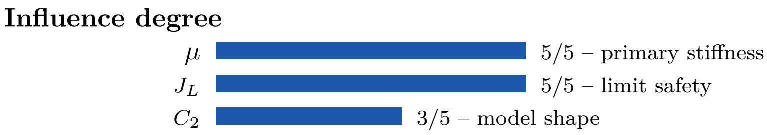

Figure 6.20: Relative influence degree of model parameters on the material response.

6.20.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.21 Model 21: Gregory

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar, simple shear

6.21.1 Strain-energy function

$$W = \frac{A}{2-n}(I_1 - 3 + C^2)^{1-n/2} + \frac{B}{2+m}(I_1 - 3 + C^2)^{1+m/2}$$

6.21.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
A	-20	20	0.05	stress	-
B	-20	20	0.05	stress	-
C	1e-06	100	1	dimensionless	-
n	-20	1.95	0	dimensionless	-
m	-1.95	20	0	dimensionless	-

Table 6.21: Model parameter search ranges.

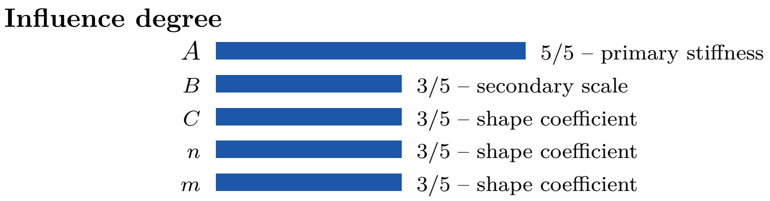

Figure 6.21: Relative influence degree of model parameters on the material response.

6.21.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.22 Model 22: Edwards-Vilgis

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar

6.22.1 Strain-energy function

$$W = \frac{\mu}{2}[\frac{(J_L + 2)(J_L - 3)(I_1 - 3)}{J_L(J_L - I_1 + 3)} + \ln(1 - \frac{I_1 - 3}{J_L})]$$

6.22.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
J_L	3.01	1000	50	değişmez limiti	-

Table 6.22: Model parameter search ranges.

Figure 6.22: Relative influence degree of model parameters on the material response.

6.22.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.23 Model 23: David De-Thomas

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar, simple shear

6.23.1 Strain-energy function

$$W = \frac{A}{2-n}(I_1 - 3 + C^2)^{1-n/2} + k(I_1 - 3)^2$$

6.23.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
A	-20	20	0.05	stress	-
C	1e-06	100	1	dimensionless	-
n	-20	1.95	0	dimensionless	-
k	-20	20	0	stress	-

Table 6.23: Model parameter search ranges.

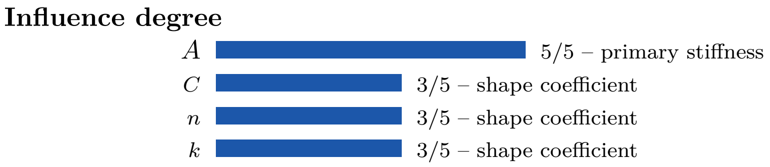

Figure 6.23: Relative influence degree of model parameters on the material response.

6.23.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.24 Model 24: Gent-Thomas

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar, simple shear

6.24.1 Strain-energy function

$$W = C_1(I_1 - 3) + 3C_2 \ln I_2$$

6.24.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_1	0	20	0.05	positive stress	-
C_2	-20	20	0	stress	-

Table 6.24: Model parameter search ranges.

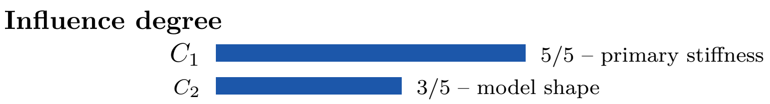

Figure 6.24: Relative influence degree of model parameters on the material response.

6.24.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.25 Model 25: Yeoh-Fleming

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar

6.25.1 Strain-energy function

$$W = \frac{A}{B}(1 - e^{-B(I_1-3)}) - C_{10}(I_L - 3)\ln(1 - \frac{I_1 - 3}{I_L - 3})$$

6.25.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
A	0	20	0.05	positive stress	-
B	1e-08	100	1	dimensionless	-
C_{10}	0	20	0.05	positive stress	-
I_L	3.01	1000	50	değişmez limiti	-

Table 6.25: Model parameter search ranges.

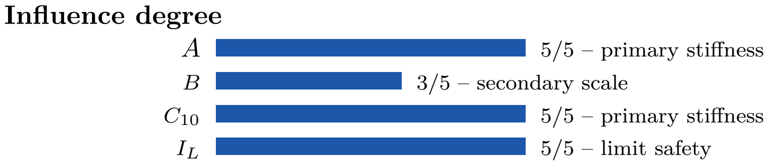

Figure 6.25: Relative influence degree of model parameters on the material response.

6.25.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.26 Model 26: 4 term H. Bechir et al.

Model summary

Model family: Principal-stretch / spectral family
Recommended data: uniaxial, biaxial, planar, simple shear

6.26.1 Strain-energy function

$$W = \sum_{n=1}^{2} \sum_{r=1}^{2} C_n^r (\lambda_1^{2n} + \lambda_2^{2n} + \lambda_3^{2n} - 3)^r$$

6.26.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{11}	-20	20	0.05	stress	-
C_{12}	-20	20	0	stress	-
C_{21}	-20	20	0	stress	-
C_{22}	-20	20	0	stress	-

Table 6.26: Model parameter search ranges.

Figure 6.26: Relative influence degree of model parameters on the material response.

6.26.3 Engineering interpretation

This model is directly driven by principal stretches. It can represent tension and compression flexibly, but stress-scale parameters and exponent parameters may compensate each other. Parameters obtained from uniaxial data only should not be treated as unique.

6.27 Model 27: 6 term H. Bechir et al.

Model summary

Model family: Principal-stretch / spectral family
Recommended data: uniaxial, biaxial, planar, simple shear

6.27.1 Strain-energy function

$$W = \sum_{n=1}^{3} \sum_{r=1}^{2} C_n^r (\lambda_1^{2n} + \lambda_2^{2n} + \lambda_3^{2n} - 3)^r$$

6.27.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{11}	-20	20	0.05	stress	-
C_{12}	-20	20	0	stress	-
C_{21}	-20	20	0	stress	-
C_{22}	-20	20	0	stress	-
C_{31}	-20	20	0	stress	-
C_{32}	-20	20	0	stress	-

Table 6.27: Model parameter search ranges.

Figure 6.27: Relative influence degree of model parameters on the material response.

6.27.3 Engineering interpretation

This model is directly driven by principal stretches. It can represent tension and compression flexibly, but stress-scale parameters and exponent parameters may compensate each other. Parameters obtained from uniaxial data only should not be treated as unique.

6.28 Model 28: 3 term Hartmann-Neff

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar

6.28.1 Strain-energy function

$$W = \alpha(I_1^3 - 3^3) + C_{10}(I_1 - 3) + C_{01}(I_2^{3/2} - 3\sqrt{3})$$

6.28.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
α	-20	20	0	stress	-
C_{10}	-20	20	0.05	stress	-
C_{01}	-20	20	0.05	stress	-

Table 6.28: Model parameter search ranges.

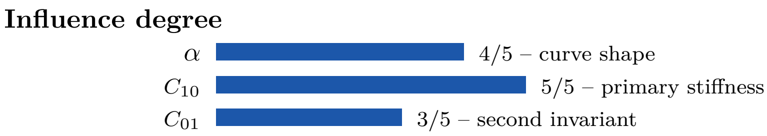

Figure 6.28: Relative influence degree of model parameters on the material response.

6.28.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.29 Model 29: 5 term Hartmann-Neff

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar, simple shear

6.29.1 Strain-energy function

$$W = \alpha(I_1^3 - 3^3) + C_{10}(I_1 - 3) + C_{01}(I_2^{3/2} - 3\sqrt{3}) + C_{20}(I_1 - 3)^2 + C_{02}(I_2^{3/2} - 3\sqrt{3})^2$$

6.29.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
α	-20	20	0	stress	-
C_{10}	-20	20	0.05	stress	-
C_{01}	-20	20	0.05	stress	-
C_{20}	-20	20	0	stress	-
C_{02}	-20	20	0	stress	-

Table 6.29: Model parameter search ranges.

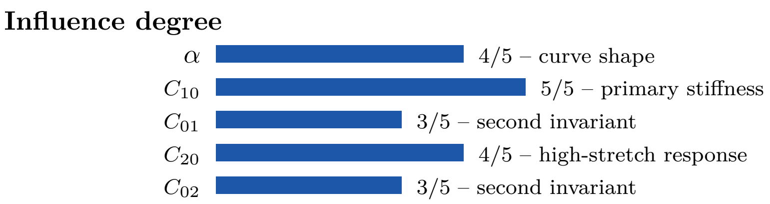

Figure 6.29: Relative influence degree of model parameters on the material response.

6.29.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.30 Model 30: 7 term Hartmann-Neff

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar, simple shear

6.30.1 Strain-energy function

$$W = \alpha(I_1^3 - 3^3) + C_{10}(I_1 - 3) + C_{01}(I_2^{3/2} - 3\sqrt{3}) + C_{20}(I_1 - 3)^2 + C_{02}(I_2^{3/2} - 3\sqrt{3})^2 + C_{30}(I_1 - 3)^3 + C_{03}(I_2^{3/2} - 3\sqrt{3})^3$$

6.30.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
α	-20	20	0	stress	-
C_{10}	-20	20	0.05	stress	-
C_{01}	-20	20	0.05	stress	-
C_{20}	-20	20	0	stress	-
C_{02}	-20	20	0	stress	-
C_{30}	-20	20	0	stress	-
C_{03}	-20	20	0	stress	-

Table 6.30: Model parameter search ranges.

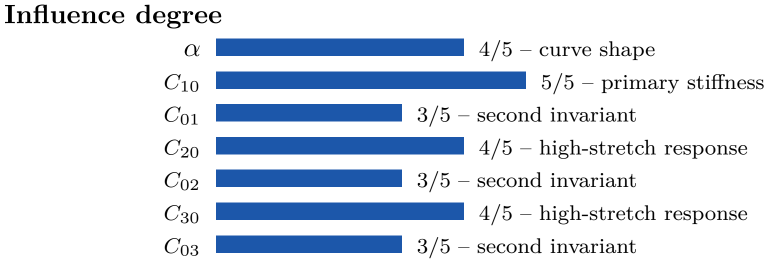

Figure 6.30: Relative influence degree of model parameters on the material response.

6.30.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.31 Model 31: Modified Yeoh

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar

6.31.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3 + \frac{\alpha}{\beta}(1 - e^{-\beta(I_1 - 3)})$$

6.31.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0	20	0.05	positive stress	-
C_{20}	-20	20	0	stress	-
C_{30}	-20	20	0	stress	-
α	0	20	0.05	positive stress	-
β	1e-08	100	1	dimensionless	-

Table 6.31: Model parameter search ranges.

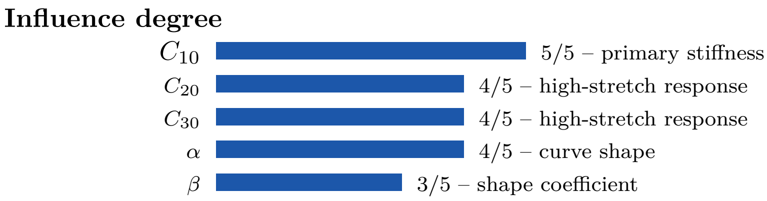

Figure 6.31: Relative influence degree of model parameters on the material response.

6.31.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.32 Model 32: Van Der Waals

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar, simple shear

6.32.1 Strain-energy function

$$W = \mu\left[-(\lambda_m^2 - 3)\left(\ln(1-\eta) + \eta\right) - \frac{2a}{3}\left(\frac{I_1 - 3}{2}\right)^{3/2}\right], \quad \eta = \sqrt{\frac{(1-\beta)I_1 + \beta I_2 - 3}{\lambda_m^2 - 3}}$$

6.32.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
λ_m	1.74	30	5.0	uzama limiti	-
a	-10	10	0.0	dimensionless	-
β	0.0	1.0	0.5	dimensionless	-

Table 6.32: Model parameter search ranges.

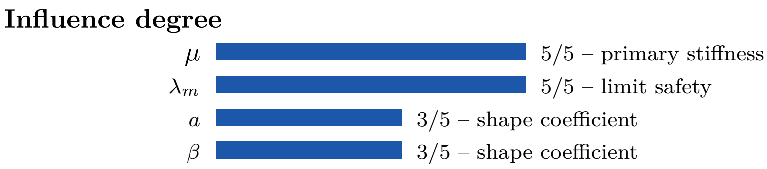

Figure 6.32: Relative influence degree of model parameters on the material response.

6.32.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.33 Model 33: Fung

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar

6.33.1 Strain-energy function

$$W = \frac{\mu}{2b}\left(e^{b(I_1-3)} - 1\right)$$

6.33.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
b	1e-08	100	1	dimensionless	-

Table 6.33: Model parameter search ranges.

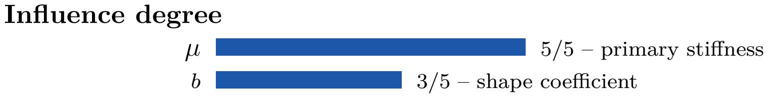

Figure 6.33: Relative influence degree of model parameters on the material response.

6.33.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.34 Model 34: Horgan-Saccomandi

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar

6.34.1 Strain-energy function

$$W = -\frac{\mu}{2} J_L \ln\left(\frac{J_L^3 - J_L^2 I_1 + J_L I_2 - 1}{(J_L - 1)^3}\right)$$

6.34.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
J_L	3.01	1000	50	değişmez limiti	-

Table 6.34: Model parameter search ranges.

Figure 6.34: Relative influence degree of model parameters on the material response.

6.34.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately.

If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.35 Model 35: Kilian

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar

6.35.1 Strain-energy function

$$W = -\mu J_L[\ln(1 - \sqrt{\frac{I_1 - 3}{J_L}}) + \sqrt{\frac{I_1 - 3}{J_L}}]$$

6.35.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
J_L	3.01	1000	50	değişmez limiti	-

Table 6.35: Model parameter search ranges.

Figure 6.35: Relative influence degree of model parameters on the material response.

6.35.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.36 Model 36: 3-parameter Gent

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar, simple shear

6.36.1 Strain-energy function

$$W = \frac{\mu}{2}[-\alpha(I_L - 3)\ln(1 - \frac{I_1 - 3}{I_L - 3}) + (1 - \alpha)(I_2 - 3)]$$

6.36.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0	20	0.1	positive stress	-
α	0	1	0.8	dimensionless	-
I_L	3.01	1000	50	değişmez limiti	-

Table 6.36: Model parameter search ranges.

Figure 6.36: Relative influence degree of model parameters on the material response.

6.36.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.37 Model 37: Low-strain Hoss-Marczak

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar

6.37.1 Strain-energy function

$$W = \frac{\alpha}{\beta}(1 - e^{-\beta(I_1-3)}) + \frac{\mu}{2b}[(1 + \frac{b(I_1-3)}{n})^n - 1]$$

6.37.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
α	0	20	0.05	positive stress	-
β	1e-08	100	1	dimensionless	-
μ	0	20	0.1	positive stress	-
b	1e-08	100	1	dimensionless	-
n	0.05	20	2	dimensionless	-

Table 6.37: Model parameter search ranges.

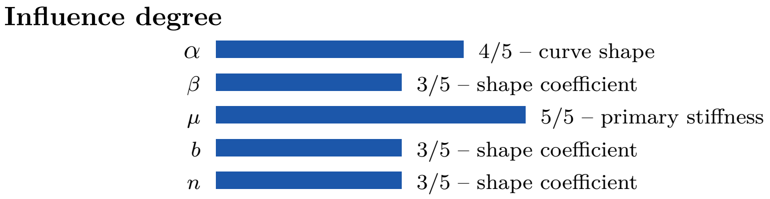

Figure 6.37: Relative influence degree of model parameters on the material response.

6.37.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.38 Model 38: High-strain Hoss-Marczak

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar, simple shear

6.38.1 Strain-energy function

$$W = \frac{\alpha}{\beta}(1 - e^{-\beta(I_1-3)}) + \frac{\mu}{2b}[(1 + \frac{b(I_1-3)}{n})^n - 1] + C_2 \ln(\frac{1}{3}I_2)$$

6.38.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
α	0	20	0.05	positive stress	-
β	1e-08	100	1	dimensionless	-
μ	0	20	0.1	positive stress	-
b	1e-08	100	1	dimensionless	-
n	0.05	20	2	dimensionless	-
C_2	-20	20	0	stress	-

Table 6.38: Model parameter search ranges.

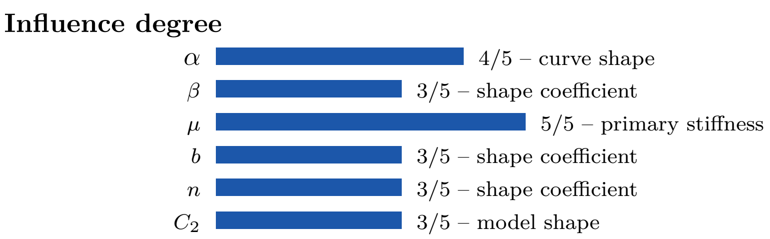

Figure 6.38: Relative influence degree of model parameters on the material response.

6.38.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.39 Model 39: Improved Hart-Smith

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar

6.39.1 Strain-energy function

$$W = \frac{C_1 e^{C_3(I_1-3)^n}}{n} + 3C_2 \ln(I_2)$$

6.39.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_1	0	20	0.05	positive stress	-
C_2	-20	20	0	stress	-
C_3	1e-08	100	1	dimensionless	-
n	0.1	10	2	dimensionless	-

Table 6.39: Model parameter search ranges.

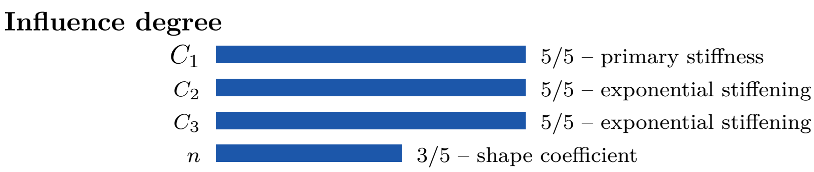

Figure 6.39: Relative influence degree of model parameters on the material response.

6.39.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.40 Model 40: Takamizawa-Hayashi

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar

6.40.1 Strain-energy function

$$W = -c \ln[1 - (\frac{I_1 - 2}{J_L})^2]$$

6.40.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
c	0	20	0.05	positive stress	-
J_L	3.01	1000	50	değişmez limiti	-

Table 6.40: Model parameter search ranges.

Figure 6.40: Relative influence degree of model parameters on the material response.

6.40.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.41 Model 41: Yamashita-Kawabata

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar

6.41.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + \frac{C_3}{N+1}(I_1 - 3)^{N+1}$$

6.41.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0	20	0.05	positive stress	-
C_3	-20	20	0	stress	-
N	0.05	10	2	dimensionless	-

Table 6.41: Model parameter search ranges.

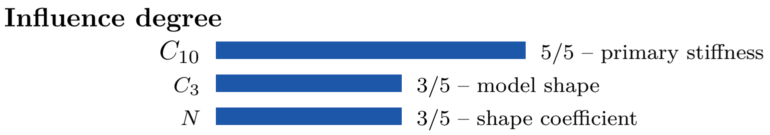

Figure 6.41: Relative influence degree of model parameters on the material response.

6.41.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.42 Model 42: Amin

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar, simple shear

6.42.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + \frac{C_3}{N+1}(I_1 - 3)^{N+1} + \frac{C_4}{M+1}(I_1 - 3)^{M+1}$$

6.42.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0	20	0.05	positive stress	-
C_3	-20	20	0	stress	-
N	0.05	10	2	dimensionless	-
C_4	-20	20	0	stress	-
M	0.05	10	3	dimensionless	-

Table 6.42: Model parameter search ranges.

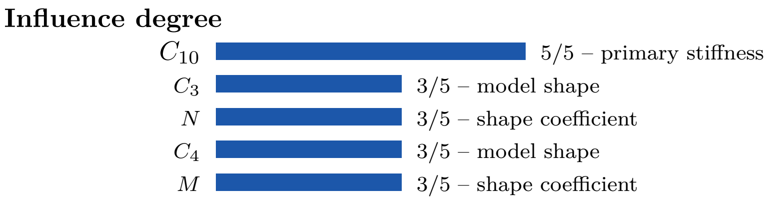

Figure 6.42: Relative influence degree of model parameters on the material response.

6.42.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.43 Model 9010: Embedded Plugin - One-term Ogden

Model summary

Model family: Principal-stretch / spectral family
Recommended data: uniaxial, biaxial, planar, simple shear

6.43.1 Strain-energy function

$$W = \frac{\mu}{\alpha}\left(\lambda_1^\alpha + \lambda_2^\alpha + \lambda_3^\alpha - 3\right)$$

6.43.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0.0	20.0	0.1	positive stress	stress
α	0.15	12.0	2.0	dimensionless	-

Table 6.43: Model parameter search ranges.

Figure 6.43: Relative influence degree of model parameters on the material response.

6.43.3 Engineering interpretation

This model is directly driven by principal stretches. It can represent tension and compression flexibly, but stress-scale parameters and exponent parameters may compensate each other. Parameters obtained from uniaxial data only should not be treated as unique.

6.44 Model 9011: Embedded Plugin - Demiray exponential

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar

6.44.1 Strain-energy function

$$W = \frac{a}{2b}\left(e^{b(I_1-3)} - 1\right)$$

6.44.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
a	0.0	20.0	0.1	positive stress	stress
b	0.0001	80.0	1.0	dimensionless	-

Table 6.44: Model parameter search ranges.

Figure 6.44: Relative influence degree of model parameters on the material response.

6.44.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.45 Model 9012: Embedded Plugin - Carroll high-strain

Model summary

Model family: Exponential stiffening family
Recommended data: uniaxial, biaxial, planar, simple shear

6.45.1 Strain-energy function

$$W = A(I_1 - 3) + B(I_1^4 - 81) + C(\sqrt{I_2} - \sqrt{3})$$

6.45.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
A	0.0	20.0	0.1	positive stress	stress
B	-2.0	2.0	0.0	stress	stress
C	-20.0	20.0	0.0	stress	stress

Table 6.45: Model parameter search ranges.

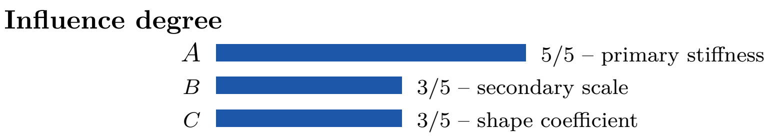

Figure 6.45: Relative influence degree of model parameters on the material response.

6.45.3 Engineering interpretation

This model contains exponential stiffening. Exponential coefficients can increase stress very rapidly at high strain. Wide upper bounds can cause numerical overflow or artificial stiffening, so extrapolation plots should always be reviewed after calibration.

6.46 Model 9013: Embedded Plugin - Lopez-Pamies two-term

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar

6.46.1 Strain-energy function

$$W = \sum_{r=1}^{2} \frac{\mu_r}{\alpha_r} \left(I_1^{\alpha_r/2} - 3^{\alpha_r/2} \right)$$

6.46.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ_1	0.0	20.0	0.1	positive stress	stress
α_1	0.2	12.0	2.0	dimensionless	-
μ_2	-20.0	20.0	0.0	stress	stress
α_2	0.2	16.0	6.0	dimensionless	-

Table 6.46: Model parameter search ranges.

Figure 6.46: Relative influence degree of model parameters on the material response.

6.46.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.47 Model 9014: Embedded Plugin - Isihara three-parameter

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar

6.47.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) + C_{20}(I_1 - 3)^2$$

6.47.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0.0	20.0	0.1	positive stress	stress
C_{01}	-20.0	20.0	0.0	stress	stress
C_{20}	-20.0	20.0	0.0	stress	stress

Table 6.47: Model parameter search ranges.

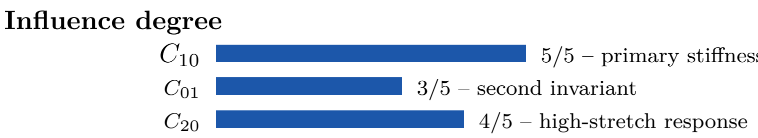

Figure 6.47: Relative influence degree of model parameters on the material response.

6.47.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.48 Model 9015: Embedded Plugin - Haines-Wilson five-parameter

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar, simple shear

6.48.1 Strain-energy function

$$W = C_{10}x + C_{01}y + C_{20}x^2 + C_{30}x^3 + C_{11}xy, \quad x = I_1 - 3,\ y = I_2 - 3$$

6.48.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0.0	20.0	0.1	positive stress	stress
C_{01}	-20.0	20.0	0.0	stress	stress
C_{20}	-20.0	20.0	0.0	stress	stress
C_{30}	-10.0	10.0	0.0	stress	stress
C_{11}	-20.0	20.0	0.0	stress	stress

Table 6.48: Model parameter search ranges.

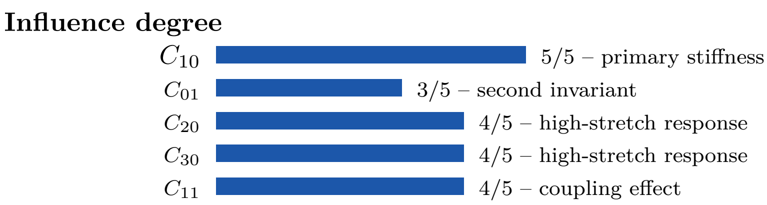

Figure 6.48: Relative influence degree of model parameters on the material response.

6.48.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.49 Model 9016: Embedded Plugin - Biderman four-parameter

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar

6.49.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3$$

6.49.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0.0	20.0	0.1	positive stress	stress
C_{01}	-20.0	20.0	0.0	stress	stress
C_{20}	-20.0	20.0	0.0	stress	stress
C_{30}	-10.0	10.0	0.0	stress	stress

Table 6.49: Model parameter search ranges.

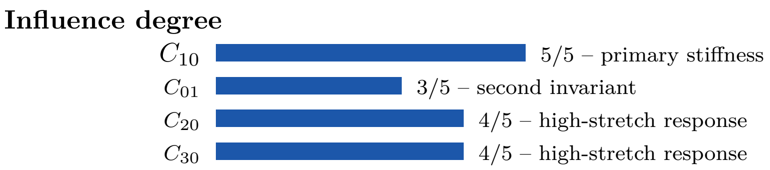

Figure 6.49: Relative influence degree of model parameters on the material response.

6.49.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.50 Model 9017: Embedded Plugin - Gent-Thomas-Yeoh hybrid

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar, simple shear

6.50.1 Strain-energy function

$$W = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2 + 3C_2 \ln\left(\frac{I_2}{3}\right)$$

6.50.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0.0	20.0	0.1	positive stress	stress
C_{20}	-20.0	20.0	0.0	stress	stress
C_2	-20.0	20.0	0.0	stress	stress

Table 6.50: Model parameter search ranges.

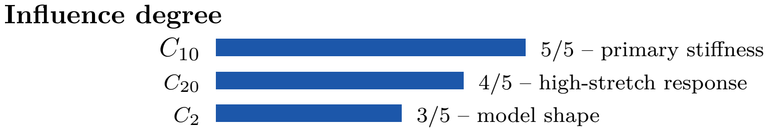

Figure 6.50: Relative influence degree of model parameters on the material response.

6.50.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.51 Model 9018: Embedded Plugin - Exponential-polinom I1/I2

Model summary

Model family: Invariant-based polynomial family
Recommended data: uniaxial, biaxial, planar

6.51.1 Strain-energy function

$$W = \frac{A}{B}\left(e^{B(I_1-3)} - 1\right) + C_{01}(I_2 - 3) + C_{20}(I_1 - 3)^2$$

6.51.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
A	0.0	20.0	0.1	positive stress	stress
B	0.0001	60.0	1.0	dimensionless	-
C_{01}	-20.0	20.0	0.0	stress	stress
C_{20}	-20.0	20.0	0.0	stress	stress

Table 6.51: Model parameter search ranges.

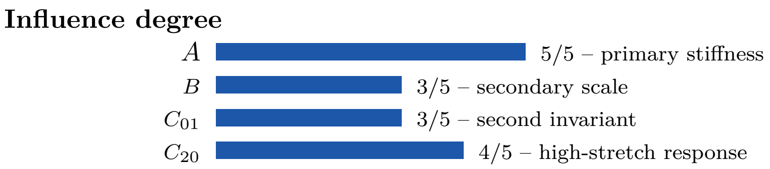

Figure 6.51: Relative influence degree of model parameters on the material response.

6.51.3 Engineering interpretation

This is an invariant-based polynomial model. Since first- and second-invariant contributions act together, multiple deformation modes are preferred. A lower residual from additional parameters does not automatically imply physical validity; slope continuity and high-stretch behavior should be checked.

6.52 Model 9019: Embedded Plugin - Shifted mixed power-law

Model summary

Model family: General phenomenological hyperelastic family
Recommended data: uniaxial, biaxial, planar, simple shear

6.52.1 Strain-energy function

$$W = \frac{A}{n}\left[(I_1 - 3 + c)^n - c^n\right] + C_{01}(I_2 - 3)$$

6.52.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
A	0.0	20.0	0.1	positive stress	stress
n	0.2	6.0	1.5	dimensionless	-
c	0.05	20.0	1.0	dimensionless	-
C_{01}	-20.0	20.0	0.0	stress	stress

Table 6.52: Model parameter search ranges.

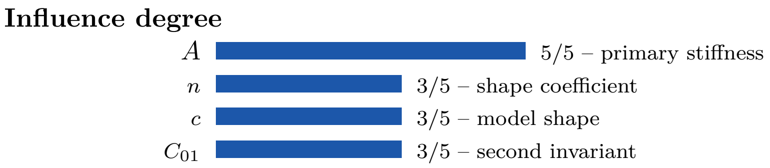

Figure 6.52: Relative influence degree of model parameters on the material response.

6.52.3 Engineering interpretation

The parameters should be considered reliable only within the deformation range covered by calibration data. Residual error, initial slope, high-stretch slope, mode-to-mode consistency and proximity to parameter bounds should be reviewed together.

6.53 Model 9020: Embedded Plugin - 4 term reduced polynomial

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar

6.53.1 Strain-energy function

$$W = C_{10}x + C_{20}x^2 + C_{30}x^3 + C_{40}x^4, \quad x = I_1 - 3$$

6.53.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
C_{10}	0.0	20.0	0.1	positive stress	stress
C_{20}	-20.0	20.0	0.0	stress	stress
C_{30}	-10.0	10.0	0.0	stress	stress
C_{40}	-5.0	5.0	0.0	stress	stress

Table 6.53: Model parameter search ranges.

Figure 6.53: Relative influence degree of model parameters on the material response.

6.53.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

6.54 Model 9021: Embedded Plugin - Gent-Yeoh finite extensibility

Model summary

Model family: Network / finite-extensibility family
Recommended data: uniaxial, biaxial, planar, simple shear

6.54.1 Strain-energy function

$$W = -\frac{\mu J_m}{2} \ln\left(1 - \frac{I_1 - 3}{J_m}\right) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3$$

6.54.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
μ	0.0	20.0	0.1	positive stress	stress
J_m	0.1	500.0	50.0	değişmez limiti	-
C_{20}	-20.0	20.0	0.0	stress	stress
C_{30}	-10.0	10.0	0.0	stress	stress

Table 6.54: Model parameter search ranges.

Figure 6.54: Relative influence degree of model parameters on the material response.

6.54.3 Engineering interpretation

The limiting parameter is not just a fitting coefficient; it controls how rapidly the material stiffens at high stretch. The largest sampled deformation and the fitted limit should be reported separately. If the limit is too close to the test range, the curve may look well fitted but will become artificially steep immediately after the last data point.

6.55 Model 9022: Embedded Plugin - Treloar I1 series

Model summary

Model family: Reduced polynomial / first-invariant family
Recommended data: uniaxial, biaxial, planar

6.55.1 Strain-energy function

$$W = A_1(I_1 - 3) + A_2(I_1^2 - 9) + A_3(I_1^3 - 27)$$

6.55.2 Parameter range and influence degree

Parameter	Lower bound	Upper bound	Initial	Effect class	Unit
A_1	0.0	20.0	0.1	positive stress	stress
A_2	-5.0	5.0	0.0	stress	stress
A_3	-2.0	2.0	0.0	stress	stress

Table 6.55: Model parameter search ranges.

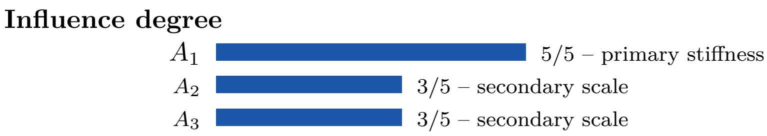

Figure 6.55: Relative influence degree of model parameters on the material response.

6.55.3 Engineering interpretation

This model emphasizes first-invariant response. It can be stable and efficient with uniaxial data, but biaxial and planar behavior is not automatically validated. High-order terms should be checked for oscillation, overfitting and nonphysical slope outside the data range.

7 Conclusion

A reliable hyperelastic model fits the data with sufficient accuracy, remains consistent across deformation modes, avoids attachment to limiting parameters and has a clearly reported validity range. Therefore, the final report should include not only the error value, but also model family, test-mode coverage, parameter influence degrees, limit ratios and extrapolation behavior.